2.2.2 The interoperability problem

The internet was built without an identity layer for humans. Consequently, every organization has solved the proximity problem in its own way. The result is far from seamless, resulting in a myriad of user experiences, hundreds of different usernames and identities, and thousands of online forms that all ask for nearly the same information.

Organizations cope with the interoperability problem using protocols like System for Cross-domain Identity Management (SCIM, https://www.okta.com/blog/2017/01/what-is-scim) for syncing information between systems and OpenID Connect (https://auth0.com/docs/protocols/oidc) for federating identity providers. If you've ever used your Google, Facebook, or Apple account to log into another app or service, you've used OpenID Connect. These and other protocols help with interoperability, but they don't solve the entire problem.

The core of this problem is that people have no identity systems of their own and, consequently, are just visitors in the identity systems of the many organizations with whom they have relationships. People working on decentralized identity or self-sovereign identity are hoping to solve this problem. This lack of standing in the digital realm and the problems it poses are beyond the scope of this book, but in chapter 16, we will look at how decentralized credentials can result in better authorization decisions.

2.2.3 The flexibility problem

Most identity systems are built to handle specific needs. The schema for the attributes they store can be limited. For example, if your organization is using Google as your identity provider, you can only use the attributes that Google chooses. If you want others, you'll have to build a system for collecting those as part of the account profile. Solutions for workforce identity usually allow their schema to be extended, but may still have limitations for many applications.

Inflexible schemas and limited attributes directly impact authorization systems, which may need to make decisions based on the context provided by whatever attributes the identity system provides. This is especially true of